PELI-CAN

PELICAN. PELI THE CAN. PELI-CAN

Presented by

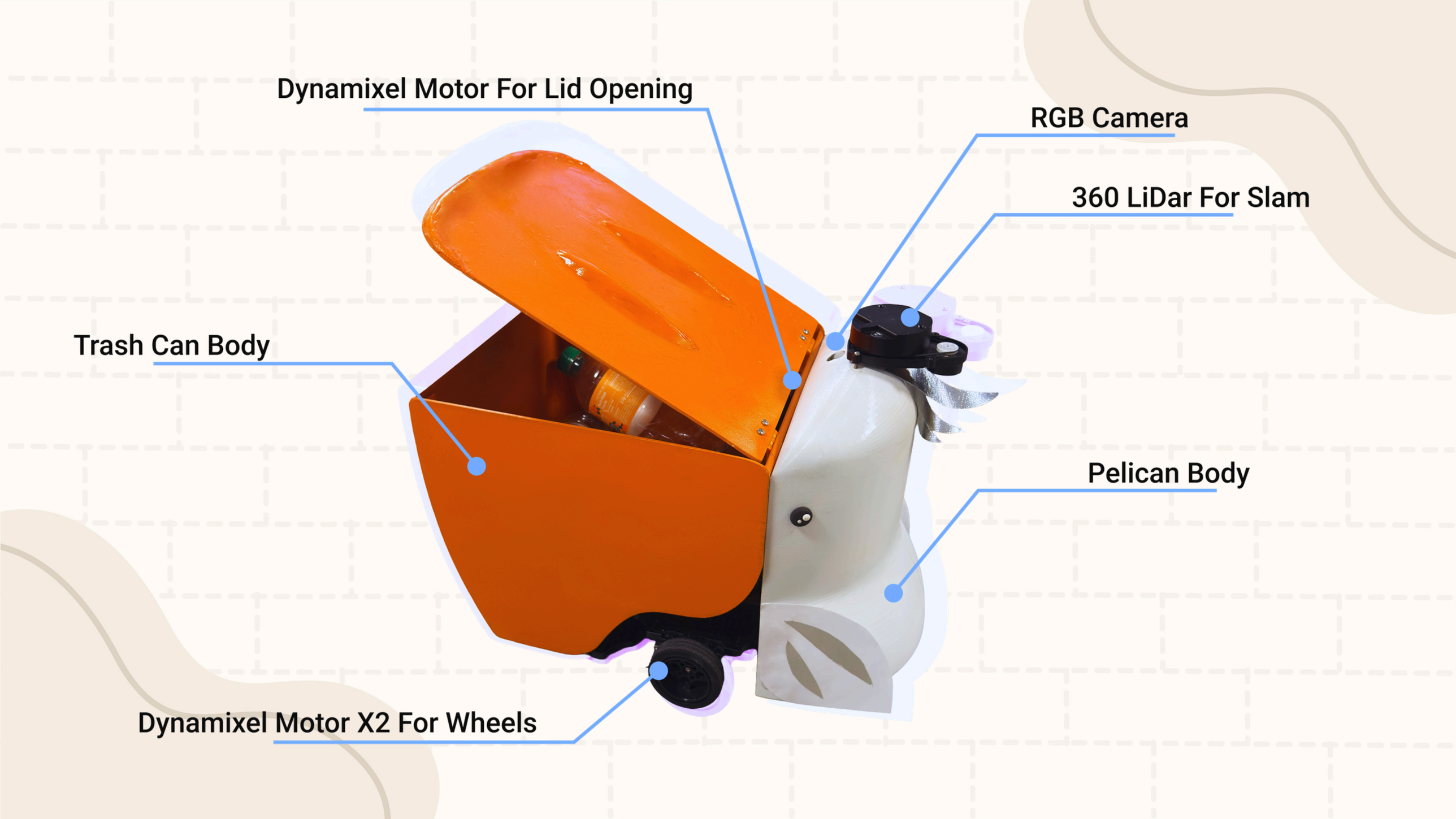
Joyce Zhou
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PROJECT OVERVIEW

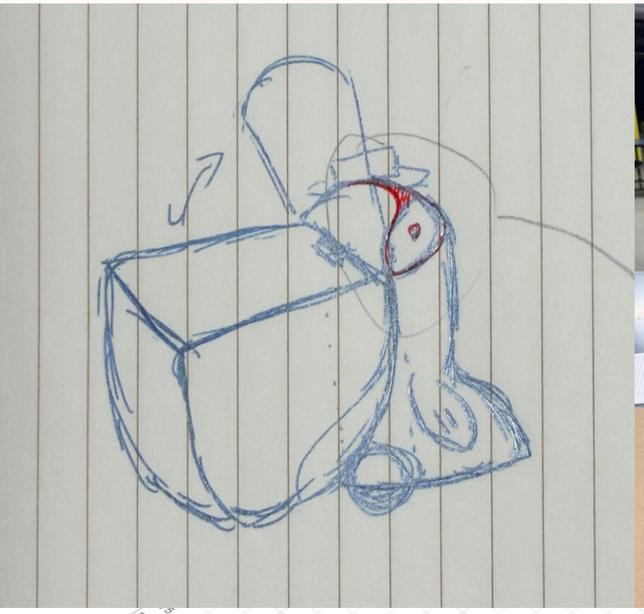


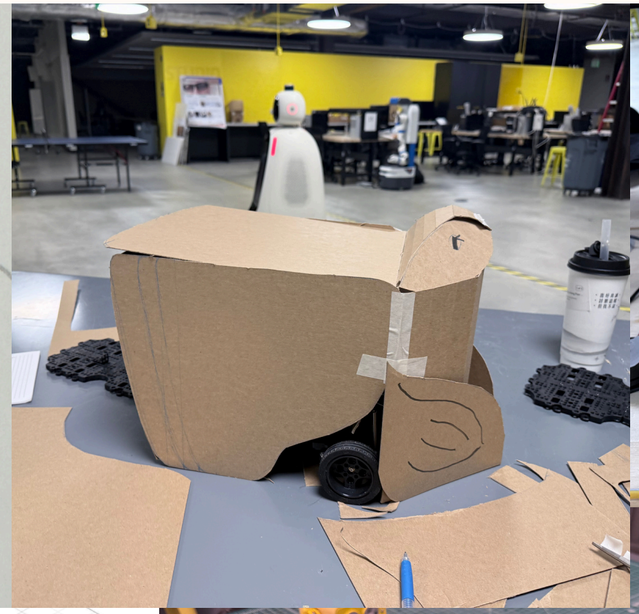
PELICAN operates as an autonomous, vision-guided system that continuously explores its environment while monitoring for disposal opportunities.

In its default state, the robot wanders around indoor spaces at a steady speed, using LiDAR to avoid obstacles. At the same time, its camera detects potential trash in real time. When a target is identified, the robot switches to tracking mode, moves toward it, and aligns the object to the center of its view. Once in position, it opens its lid—like a pelican's beak—to receive the trash.


Dynamixel Motor For Lid Opening
RGB Camera
360 LiDar For Slam
Trash Can Body
Pelican Body
Dynamixel Motor X2 For Wheels

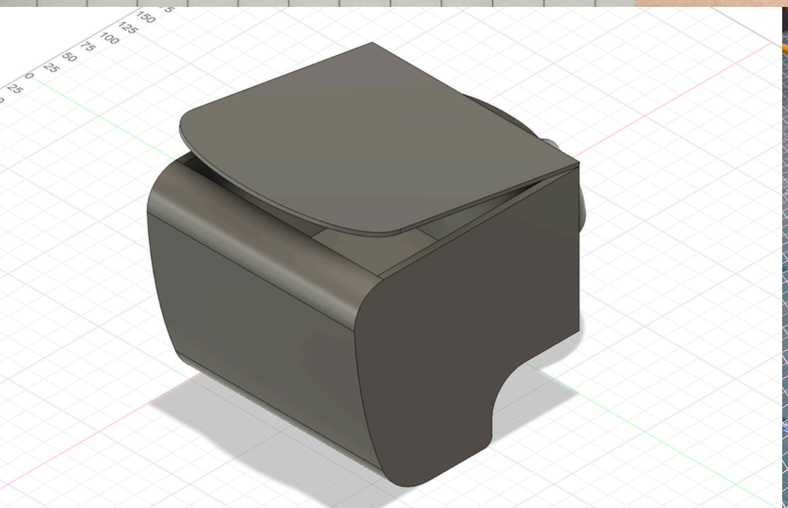
PROCESS WORK

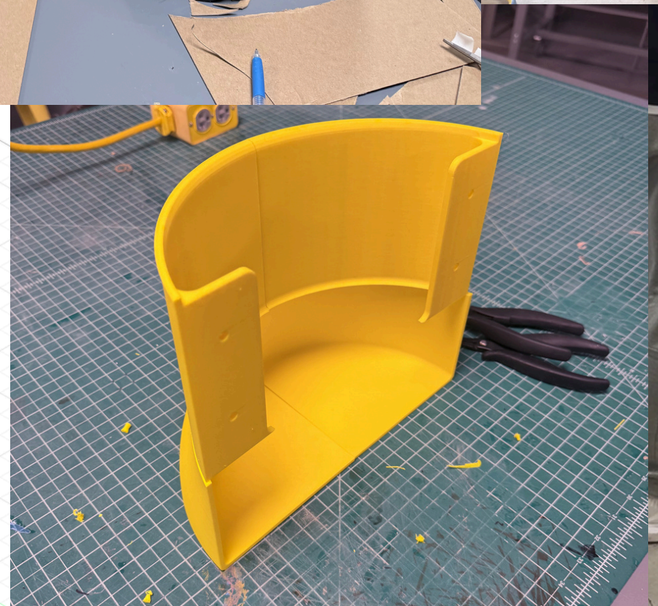
















TECHNICAL OVERVIEW

- Detects trash using YOLO bounding boxes (/yolo/detections)
- Tracks target using visual alignment (PID control)
- Publishes movement via /cmd_vel to approach object
- Uses LiDAR (/scan) for obstacle avoidance + wandering
- Switches between WANDER ↔ TRACK states
- Triggers lid motor when target is aligned



IMPROVEMENTS & FUTURE PLANS



- **Increase Capacity**
  - Enable the system to carry and manage a larger volume of waste
- **Audio Feedback**
  - Add sound or voice cues to improve user interaction and system clarity
- **Extended Perception Range**
  - Detect trash beyond the current camera view using wide-angle lenses or multi-camera setup
- **Improve Detection Accuracy**
  - Expand training data to recognize more object types with higher reliability
- **Faster Response Time**
  - Optimize navigation and control for quicker movement and reduced latency

THANK YOU

AND...

